IMPERIAL

Department of Computing
70015: Mathematics for Machine Learning
Lecturer: Matthew Wicker

Practice Problems 00: Review and Warm-up

Probability Theory

Question 1 (Set Theory and Probability)**.** Using the three axioms of probability show that

a. Write down the sample space of a dice. In your notation, use the set A to denote the event of a 3 or 4 occurring. What is the complement of A, denoted $\neg A$?

b. For a problem about lengths, we have a sample space $\Omega = [0, 1]$. For $A = (0.3, 0.4]$, what is $\neg A$?

c. $\mathbb{P}(\neg A) = 1 - \mathbb{P}(A)$

d. $\mathbb{P}(\varnothing) = 0$, where $\varnothing$ is the empty set

e. $0 \leq \mathbb{P}(A) \leq 1$

f. $A \subseteq B \implies \mathbb{P}(A) \leq \mathbb{P}(B)$

Hint: Consider the following definition. $B \backslash A = \{x \in B : x \notin A\}$

g. $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$

h. (*) if $\{A_i\}_{i=1}^{\infty} \subseteq \Omega$ and $A_i \subseteq A_{i+1} \forall i$ then:

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{i \to \infty} \mathbb{P}(A_i)$$

Hint: Use axiom 3. *: The emphasis of this course isn't on these kinds of details, even though this should be doable with 1st-year calculus.

i. For two mutually exclusive events A, B, what is $\mathbb{P}(A \cup B)$?

Question 2 (Independent events)**. Independent events don't come up as much as independent random variables, so it's ok to just follow this answer, rather than spending lots of time on it.** When tossing two coins (where we care about the order), we have a sample space $\Omega = \{HH, HT, TH, TT\}$.

a. What outcomes are contained in the event that corresponds to the the first coin being heads? We denote the event E_{1H}, and others similarly.

b. If you assume that all outcomes have equal probability, show that E_{1H} and E_{2T} are independent.

c. If you assume that E_{1H} and E_{2H} are independent and 0.5 each, show that all outomes must have equal probability.

Question 3 (Independent events)**. Independent events don't come up as much as independent random variables, so it's ok to just follow this answer, rather than spending lots of time on it.** When tossing two coins (where we care about the order), we have a sample space $\Omega = \{HH, HT, TH, TT\}$.

a. What outcomes are contained in the event that corresponds to the the first coin being heads? We denote the event E_{1H}, and others similarly.

b. If you assume that all outcomes have equal probability, show that E_{1H} and E_{2T} are independent.

c. If you assume that E_{1H} and E_{2H} are independent and 0.5 each, show that all outomes must have equal probability.

Question 4 (Random Variables)**.** Consider throwing two fair dice.

a. What is the sample space for all outcomes that you can get from throwing two dice? We specify the probability of each outcome to be the same.

b. Define two random variables A, B which map the outcome to the face value on each die respectively. Find the probability mass function for A from the probability on outcomes. The answer will work from the definition of a random variable, but you will probably intuitively get the right answer as well.

c. Show that A and B are independent.

d. Define the random variable $C = A + B$. Derive the probability mass function of C.

Question 5 (Continuous Random Variables)**.** Consider the random variable X with a probability density $p(x) = C \cdot x$ when $x \in [0, 1]$ and 0 elsewhere.

a. Calculate C.

b. Calculate $\mathbb{P}(0.3 \leq X \leq 0.75)$.

c. Calculate $\mathbb{P}(X \in [0.3, 0.75] \cup [0.8, 0.9])$.

d. Calculate $\mathbb{E}_X[X]$, $\mathbb{E}_X[X^2]$, $\mathbb{V}_X[X]$.

Check your answers by performing numerical integration, e.g. in Python.

Question 6 (Joint Discrete Random Variables)**.** Consider two random variables A, C, where A is the outcome of one die, and C gives the sum of A and the sum of another die B.

a. From intuition, write a table of $\mathbb{P}(C = c|A = a)$, which we use to denote the probability of C taking the value c, if we know that A has taken the value a.

b. Write a table of $\mathbb{P}(C = c, A = a)$. To help you think it through, consider a tree of outcomes that can occur. This helps illustrate independence between outcomes, which helps you figure out when you can multiply probabilities.

c. From the values in the table $\mathbb{P}(C = c, A = a)$ find $\mathbb{P}(2 \leq C \leq 4)$ and $\mathbb{P}(2 \leq C \leq 4, 2 \leq A \leq 4)$.

We will cover conditional probability more later, but for now just think it through.

Question 7 (Multivariate Integration)**.** Consider two continuous random variables X, Y with joint density $p(x, y) = C \cdot (x^2 + xy)$ when $x \in [0, 1]$ and $y \in [0, 1]$, and 0 elsewhere.

a. Find C.

b. Find $\mathbb{P}(0.3 \leq X \leq 0.5)$.

c. Find $\mathbb{P}(X < Y)$. Perform the integration twice in both orders, once integrating over x first, once by integrating over y first.

d. **Bonus:** Convince yourself that you know how to do this for $p(x, y, z) = C \cdot (x^2 + xyz)$ as well.

Check your answers by performing numerical integration, e.g. in Python.

Question 8 (Statistics Terminology)**.** Recall the following statistical terminology.

a. What is a statistic?

b. What is an estimator?

c. What is a consistent estimator?

d. What is a sample?

Question 9 (Vector notation)**.** We define the probability density on the vector $\mathbf{x} \in \mathbb{R}^3$ with all elements $0 \leq x_k \leq 1$ as

$$p(\mathbf{x}) = \frac{1}{C}(x_1^2 + x_1x_2 + x_2^2 + 2x_2x_3)\,. \tag{10}$$

Put this into notation that only uses $\mathbf{x}$ as a single whole vector.

Linear Algebra

Question 10 (Dot product)**.** Compute $\boldsymbol{x}^\top \boldsymbol{y}$ where $\boldsymbol{x} = (1, -2, 5, -1)^\top$ and $\boldsymbol{y} = (0, 4, -3, 7)^\top$.

Question 11 (Matrix product)**.** Compute $\boldsymbol{y} = A\boldsymbol{x}$ as well as the ℓ_2 norm of $\boldsymbol{x}$ and $\boldsymbol{y}$, where

$$A = \begin{pmatrix} -1 & 4 & 7 & 2 \\ 3 & -2 & -1 & 0 \\ 5 & 3 & 0 & -1 \end{pmatrix}, \quad \boldsymbol{x} = (-3, 2, 1, 3)^\top.$$

Question 12 (Basis)**.** Which of the following set of vectors are basis for $\mathbb{R}^2$?

1. $\{(1, 1), (1, 0)\}$
2. $\{(2, 4), (3, -1)\}$
3. $\{(1, -1), (0, 2), (2, 1)\}$
4. $\{(2, -1), (-2, 1)\}$
5. $\{(0, 3)\}$

Question 13 (Span of vectors)**.** Which of the following points are within the span of $\{(-1, 0, 2), (3, 1, 0)\}$?

1. $(0, 1, 1)$
2. $(1, 1, 4)$
3. $(2, 1, 1)$
4. $(-3, 4, 2)$
5. $(0, 0, 0)$

Question 14 (Rotation matrix in $\mathbb{R}^2$)**.** What is the 2×2 matrix that rotates all the non-zero vectors in $\mathbb{R}^2$ by $45°$ counter-clockwise?

Question 15 (Linear equations)**.** Given the following system of linear equations:

$$\begin{aligned} x + 2y &= 2 \\ 3x + 2y + 4z &= 5 \\ -2x + y - 2z &= -1 \end{aligned}$$

Answer the following questions:

a Writing this system in a matrix form $A\boldsymbol{x} = \mathbf{b}$ with $\boldsymbol{x} = (x, y, z)^\top$. What are A and $\mathbf{b}$?

b Solve this system, or show that the solution does not exist.

c What is the rank of A?

Question 16 (Eigen decomposition)**.** Consider a matrix $A \in \mathbb{R}^{d\times d}$ and assume it has an eigen decomposition of $A = Q\Lambda Q^{-1}$ where $\Lambda = \text{diag}(\lambda_1, ..., \lambda_d)$. When A is symmetric we also have $Q^{-1} = Q^\top$. Answer the following questions:

a. If A is symmetric, show that $\boldsymbol{x}^\top A\boldsymbol{x} \geq 0$ for any $\boldsymbol{x} \in \mathbb{R}^{d\times 1}$ if and only if $\lambda_i \geq 0$ for all $i = 1, ..., d$.

b. Show that $Tr(A) = \sum_{i=1}^d \lambda_i$ where $Tr(A)$ is the trace of A.

c. Show that $det(A) = \prod_{i=1}^d \lambda_i$ where $det(A)$ is the determinant of A.

d. Why an entry λ_i in the diagonal matrix Λ is one of the solutions for the equation $A\boldsymbol{q} = \lambda\boldsymbol{q}$, $\boldsymbol{q} \neq \mathbf{0}$?

Calculus

Question 17 (MML 5.1-5.3)**.** This is revision. Compute the derivatives for w.r.t. x for

a. $f(x) = \log(x^4)\sin(x^3)$

b. $f(x) = (1 + \exp(-x))^{-1}$

c. $f(x) = \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$